

Applied Data Science and AI in Pharma Commercial

Fall 2026

Time: Mondays, 7:00 PM – 9:30 PM

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Office Hours: Mondays 5:00pm-6:45pm

Format: In-person lecture, guest lectures, semester-long project

Course Description

This course bridges the gap between advanced data science methodologies and their strategic application within the complex pharmaceutical and life sciences ecosystem. The commercialization of healthcare products is governed by a unique, multi-stakeholder landscape involving payers, providers, and patients.

This course is designed to prepare students for data science roles in commercial pharmaceutical and healthcare analytics. The commercial side of healthcare — manufacturers, consulting firms, payers, and the analytics vendors that serve them — is interested in data scientists who can combine modern data science and AI skills with fluency in the industry's specialized vocabulary, datasets, and decision frameworks.

By the end of the semester, students will be prepared to tackle interviews at a pharma manufacturer, a life-sciences consulting firm, or an analytics vendor and speak credibly about claims data, marketing mix modeling, HCP segmentation, omnichannel orchestration, and the agentic-AI use cases reshaping the field. They will have built end-to-end projects that they can show to a potential hiring manager.

The second deliberate goal of the course is direct, sustained exposure to industry experts. Guest lectures from senior practitioners bring real case studies into the classroom. Beyond that, every team is paired with an industry mentor — a senior commercial-analytics practitioner from a manufacturer or consulting firm — for the entire semester. Students meet their mentor on a structured cadence, defend their project framing, receive feedback on their analytical choices, and present their final work directly to mentors.

The technical content reflects what is currently being deployed in the industry. Through a blend of lectures, insights from industry-leading guest speakers, and a semester-long term project, students will learn to translate raw healthcare data into actionable, high-impact commercial strategy recommendations.

Learning Objectives

By the end of this course, students will be able to:

- Communicate to Business Stakeholders — Translate model output into clear, defensible recommendations for non-technical commercial audiences (brand, marketing, sales leadership).
- Map the relationships and incentives between pharma manufacturers, payers, providers, and patients using real-world clinical and commercial datasets.
- Apply statistical and machine learning frameworks to build Marketing Mix Models and HCP/patient segmentation strategies.
- Design Next Best Action (NBA) engines that dynamically optimize marketing interventions and reduce physician fatigue.
- Use Foundation Models and modern NLP to extract commercial signals from unstructured data such as field notes and medical literature.

- Brainstorm/design Agentic Solutions — Evaluate the architectural shift toward autonomous, agentic AI systems in life sciences commercial operations and judge where they create durable value.

Prerequisites

- Elements of Data Science (or equivalent introductory statistics + ML exposure)
- Applied Machine Learning (or equivalent supervised/unsupervised methods coursework)
- Working familiarity with Python (pandas, scikit-learn, Jupyter)

No prior knowledge of the pharmaceutical or healthcare industry is required — the first two weeks are dedicated to building domain context.

Recommended Texts and Resources

- Hastie, Tibshirani, Friedman — The Elements of Statistical Learning (selected chapters; free PDF).
- PMSA — The Pharmaceutical Management Science Association is the de facto industry venue for commercial analytics. The PMSA Journal (JPMSA), the annual conference archive, and the webinar archive are referenced throughout the weekly outline — provided as PDFs/links
- Industry primers and white papers (IQVIA, Komodo Health, Axtria, ZS, Veeva, MathCo, McKinsey, Deloitte) distributed weekly via the course site.
- Selected peer-reviewed papers on Marketing Mix Modeling, uplift modeling, and foundation models for time series — provided as PDFs.

*All assigned readings are free or distributed through the course site. **There is no required purchased textbook.***

Assignments and Grading

The midterm and final presentations will be in class. Homework assignments are take-home and due two weeks after they are released. Late submissions are not accepted.

Component	Weight
Guest-speaker engagement (questions submitted in advance + 1-page summary report; 3 speakers, 5% each)	15%
Homework assignments (two, equally weighted at 10% each)	20%
Midterm exam	25%
Term project	40%
Total	100%

Grade Scale

Quality of Performance	Letter Grade	Range %	GPA / Quality Pts.
Excellent — work is of exceptional quality	A+	99 – 100	4.33
Excellent — work is of exceptional quality	A	93 – 98.99	4.00
Excellent — work is of exceptional quality	A–	90 – 92.99	3.67

Good — work is above average	B+	87 – 89.99	3.33
Satisfactory	B	83 – 86.99	3.00
Below average	B–	80 – 82.99	2.67
Poor	C+	77 – 79.99	2.33
Poor	C	73 – 76.99	2.00
Poor	C–	70 – 72.99	1.67
Poor	D	65 – 69.99	1.00
Poor	D–	60 – 64.99	0.67
Failure	F	< 60	0.00

Term Project - Mentor-Supervised, Semester-Long

The term project is the centerpiece of the course. Students will work in teams of 3–4 on a realistic pharma commercial problem of their choosing and deliver an end-to-end analytical solution: framing, modeling, evaluation, and recommendation.

Industry Mentor

Each team will be paired with a dedicated industry mentor — a senior practitioner from either a pharmaceutical manufacturer (commercial analytics, brand, marketing, market access, or medical affairs) or a leading pharma consulting firm. Mentors are recruited from the instructor's professional network and serve as the team's principal industry sounding board for the semester.

The mentor's role is to:

- Help shape the problem statement so that it reflects an authentic commercial question.
- Sanity-check the team's data plan, modeling choices, and stakeholder framing.
- Surface industry context the textbooks miss — regulatory friction, organizational politics, market dynamics.
- Provide structured feedback at scheduled checkpoints and review the final presentation.

Mentors will not write code or do the analysis. Their job is to ensure the team's output would be credible inside a real pharma organization.

Cadence

- Mentors are introduced in Week 2 (Sep 21). Teams meet their mentor by Week 3.
- Suggested meeting cadence: 30–45 minutes every two weeks (six touchpoints over the semester).
- Mentor schedules will be coordinated outside class time. Teams are responsible for scheduling, agendas, and follow-ups.
- Two in-class mid-semester progress presentation gives the cohort visibility into each team's direction; mentors are invited.
- Final presentations (Dec 14) are attended in-person or remotely by mentors, who join the instructor in evaluating the deliverables.

Project Archetypes

Acceptable project archetypes include (but are not limited to):

- HCP segmentation + Next Best Action engine on a synthetic claims/Rx panel.
- Marketing Mix Model with calibration to a known brand launch curve.

- Patient-finding / patient-journey model using sequence models on synthetic claims.
- LLM/NLP pipeline that extracts commercial signal from medical-affairs field notes or social listening.
- Agentic workflow that automates a recurring commercial task (e.g., weekly brand performance briefing).
- More use cases to be shared in-class in week 1.

Deliverables

- A 20-minute in-class final presentation (+5 minutes Q&A) with mentor and instructor in the room.
- A code repository (notebook(s) plus README) that any teammate could clone and re-run.
- A 2-4 page executive memo summarizing problem, approach, results, recommendation, and limitations.

Weekly Outline

Week 1 — Sep 14

Introduction to the Pharmaceuticals Landscape and Terminology

- Articulate why pharma commercial analytics is structurally different.
- Identify the major stakeholders (manufacturer, payer, provider, patient, PBM, GPO, wholesaler) and their economic incentives.

Suggested readings

- IQVIA Institute — "The Use of Medicines in the U.S." (latest annual report, executive summary).

Week 2 — Sep 21

Pharma Datasets & Introduction to Claims Analytics

- Distinguish medical claims, pharmacy claims, EHR, and syndicated panel data.
- Build a basic patient cohort from synthetic claims using diagnosis and procedure codes.
- Identify common data-quality and reidentification risks in real-world data (RWD).
- Rx panel data, longitudinal vs. cross-sectional, patient capture and projection.
- Syndicated reports (DDD, Xponent-style projections) and when to trust them.

Suggested readings

- Selected sections of "Real-World Evidence Generation" (FDA guidance, executive summary).

Week 3 — Sep 28

Marketing & Media Mix Modeling and ROI Optimization

- Specify a Marketing Mix Model with adstock, saturation, and base/incremental decomposition.
- Translate a fitted MMM into channel-level ROI and a defensible reallocation recommendation.
- Recognize identifiability, multicollinearity, and seasonality pitfalls in MMM.
- Linear and Bayesian MMM; adstock (geometric/Weibull) and Hill saturation.

Suggested readings

- Jin et al., "Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects." (Google paper).
- PyMC-Marketing MMM documentation and example notebook.

Homework 1 released (due in 2 weeks).

Week 4 — Oct 3

HCP and Patient Segmentation

- Design a behavioral segmentation that marketers can actually act on.
- Compare unsupervised vs. supervised segmentation.

- Evaluate segment stability, separation, and commercial interpretability.
- Feature engineering from claims/Rx (writing behavior, patient mix, switching, channel response).

Suggested readings

- ZS / IQVIA white papers on HCP segmentation (provided).
- Selected chapter on clustering from Murphy, PML.

Week 5 — Oct 12

Omnichannel Optimization, Patient Predictive Modeling, and Next Best Action

- Frame Next Best Action as a data science problem.
- Build an HCP affinity model and a simple NBA scoring layer.
- Discuss how to manage physician fatigue, frequency caps, and channel substitution.

Suggested readings

- Veeva / Salesforce Marketing Cloud whitepapers on NBA orchestration.

Week 6 — Oct 19

Forecasting in Pharma — Epidemiology based models, Time Series, and Foundation Models

- Build an epi-based patient-flow forecast (incidence → diagnosed → treated → on-brand).
- Compare classical time-series methods (ETS, ARIMA, Prophet) to modern foundation models (TimeGPT, Chronos, Lag-Llama).
- Choose the right forecasting paradigm for launch, in-line, and loss-of-exclusivity contexts.
- Bass diffusion and analog-based launch forecasting.
- Hierarchical reconciliation across geography / specialty / channel.
- Foundation models for time series: zero-shot vs. fine-tuned.

Suggested readings

- TBD

Homework 2 released (due in 2 weeks).

Week 7 — Oct 26

NLP, Sentiment Analysis, and Topic Modeling in Pharma

- Apply modern NLP (transformers, embeddings) to pharma-specific unstructured text.
- Build a sentiment / topic pipeline on field-force notes or social listening data.
- Embeddings, retrieval, and semantic search in medical contexts.
- Adverse-event detection and domain models (BioBERT, ClinicalBERT, MedPaLM family); when generic models are good.

Suggested readings

- Lee et al., "BioBERT."
- Select FDA guidance on social-media pharmacovigilance.

Week 8 — Nov 2nd

Academic Holiday

Week 9 — Nov 9

Midterm

Week 10 — Nov 16

Guest Speaker 1 — Agentic AI Use Cases in Pharma Commercial

Week 10 — Nov 23***Term Project — Progress Check-In #1***

Each team presents 5-10 minutes + 5 minutes Q&A/feedback.

Week 11 — Nov 30***Guest Speaker 2 — The Journey of Marketing Mix in Pharma: From Regression to Agents*****Week 12 — Dec 7*****Term Project — Progress Check-In #2***

Each team presents 10-15 minutes + 10 minutes Q&A/feedback.

Week 13 — Dec 9***Panel of industry experts & Students Final Presentations***

- Students engage with a panel of industry executives.
- Students deliver a stakeholder-grade final presentation.
 - End-to-end framing → data → model → recommendation.
- 20-minute presentation + 5-minute Q&A per team. Mentors join in person or remotely as judges alongside the instructor.